

EXPERIMENT: 34 IMPLEMENTING THE SIMULATION OF ERROR CORRECTION CODE – CRC IN JAVA/C

```
#include <stdio.h>

#include <string.h>

int main()
{
    char data[100], key[50], temp[150];
    int i, j, n, keyLen;

    printf("CRC ERROR DETECTION\n");
    printf("-----\n");

    printf("Enter data bits: ");
    scanf("%s", data);

    printf("Enter generator polynomial: ");
    scanf("%s", key);

    n = strlen(data);
    keyLen = strlen(key);

    /* Append zeros */
    strcpy(temp, data);

    for(i = 0; i < keyLen - 1; i++)
```

```
strcat(temp, "0");
```

```
printf("\nData after adding zeros: %s\n", temp);
```

```
/* CRC calculation */
```

```
for(i = 0; i <= strlen(temp) - keyLen; i++)
```

```
{
```

```
    if(temp[i] == '1')
```

```
    {
```

```
        for(j = 0; j < keyLen; j++)
```

```
        {
```

```
            if(temp[i + j] == key[j])
```

```
                temp[i + j] = '0';
```

```
            else
```

```
                temp[i + j] = '1';
```

```
        }
```

```
    }
```

```
}
```

```
printf("CRC: ");
```

```
for(i = n; i < n + keyLen - 1; i++)
```

```
    printf("%c", temp[i]);
```

```
printf("\n");
```

```
printf("Transmitted Data: %s", data);
```

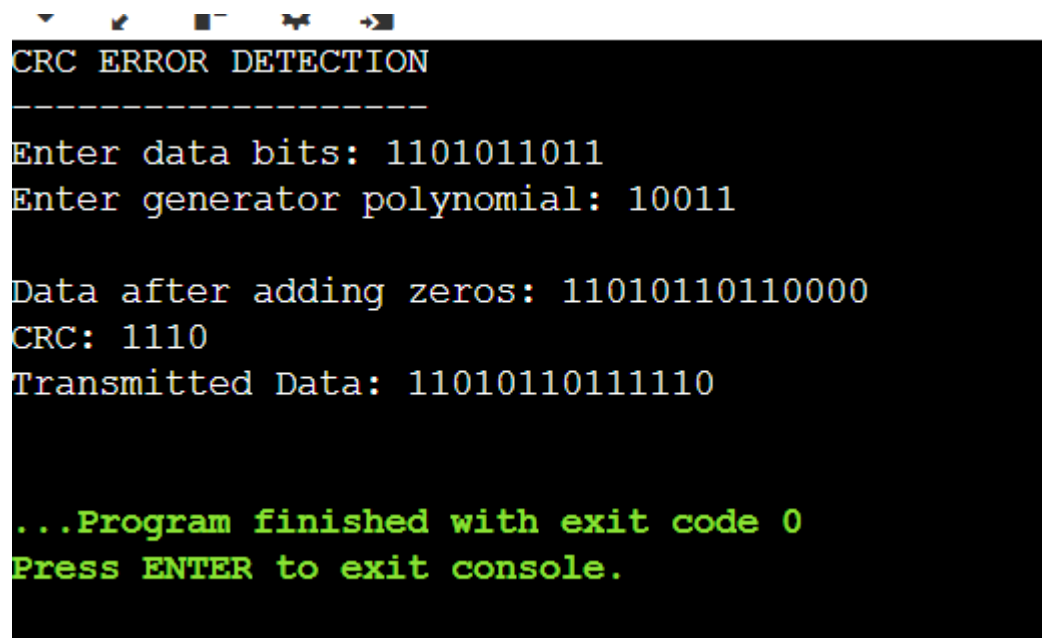
```
for(i = n; i < n + keyLen - 1; i++)
```

```
    printf("%c", temp[i]);
```

```
printf("\n");
```

```
return 0;
```

```
}
```



```
▼  ↩  ▢  ↶  ➡
```

```
CRC ERROR DETECTION
-----
Enter data bits: 1101011011
Enter generator polynomial: 10011

Data after adding zeros: 11010110110000
CRC: 1110
Transmitted Data: 11010110111110

...Program finished with exit code 0
Press ENTER to exit console.
```